

DEXTER FERNANDES

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SUMMARY

Robotics software engineer with an MSc in Robotics (Merit) and 3+ years of commercial C++ and Python experience building and supporting software on deployed industrial hardware. Built a real-time LiDAR SLAM system in C++17 and ROS 2 on a Clearpath Husky, integrating Velodyne LiDAR, RealSense depth and IMU on a Jetson AGX Xavier. Comfortable in a Linux environment with Git, Docker and CI, writing gtest and pytest coverage, and diagnosing faults that cross software, sensors and physical hardware.

RELEVANT EXPERIENCE

Esbaar – AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Hardened deployed edge services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Developed and maintained NVIDIA DeepStream and FFmpeg video pipelines in Python for 24/7 perception on BP drilling rigs, covering camera ingestion, inference and field diagnostics in desert conditions.
- Built a watchdog service that detected and restarted failed processes on edge devices, and designed its REST API so operators could act on service health and restarts remotely.
- Led on-site validation on BP rig deployments, diagnosing failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Introduced pytest, unittest and linting into CI for edge inference modules, improving release consistency across hardware targets.
- Added observability with structured logs, metrics, timeouts, retries and message validation, cutting root-cause analysis time on live deployments.
- Documented failure modes, recovery procedures and deployment guardrails for engineers supporting fielded systems, and ran peer code reviews within each Agile sprint.
- Owned the perception model lifecycle end to end, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.

LivNSense Technologies – Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, overlapping preprocessing, inference and post-processing to raise throughput from 5 to 15 FPS under real-time constraints.
- Wrote gtest unit tests and documented interfaces for the C++ inference path, catching regressions before deployment.
- Integrated Boost C++ libraries into production pipelines, resolving dependency and build issues to keep deployments reproducible.
- Profiled C++ inference and pre/post-processing hotspots with Intel VTune, targeting fixes in the CNN inference pipeline.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Tuned camera and ISP settings to keep models reliable on RGB and IR streams across 24/7 operation, cutting false positives on IR and reducing manual recalibration.
- Implemented optical flow based ego-motion estimation feeding CNN classifiers in PyTorch to infer vehicle state, supporting collision avoidance on moving vehicles.
- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.

PROJECTS

ROS 2 SLAM and Nav2 Simulation

02/2026

- Developed and evaluated a simulation-based robotics stack in ROS 2 Jazzy and Gazebo Harmonic, writing ROS 2 nodes, launch files and parameterised configs in C++ and Python.
- Used tf2, message synchronisation and timestamp alignment to debug transform timing issues, and configured Nav2 for autonomous navigation in structured scenarios.
- Added an EKF localisation layer with robot_localization fusing wheel odometry and IMU, and implemented point cloud filtering, segmentation and registration with PCL and Open3D.
- Containerised the stack with Docker on Linux and managed versioning in Git for reproducible builds across development environments.

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.

EDUCATION

University of Bristol – MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS 2 (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, validating sensor placement for coverage.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier under JetPack, working within the platform's compute and memory limits for on-robot operation.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol, configuring Ceres, g2o and GTSAM as solver backends and tuning ICP parameters.

Visvesvaraya National Institute of Technology – BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	C++17, modern C++, Python, CUDA C++, Bash, SQL
Robotics	ROS 2 (Robot Operating System 2), rclcpp, rclpy, tf2, rosbag2, launch files, Nav2, robot_localization, EKF, Gazebo, LiDAR SLAM, localisation, mobile robotics, RTAB-Map, LIO-SAM, Cartographer, GLIM, ICP (Iterative Closest Point), scan matching
Sensors	Velodyne VLP-16 LiDAR, Intel RealSense D435i depth, IMU, RGB and IR cameras, sensor integration, extrinsics calibration, multi-camera synchronisation, ISP and camera tuning, RTSP
Point clouds	PCL (Point Cloud Library), Open3D, Eigen, Ceres, g2o, GTSAM, pose graph optimisation, filtering, segmentation, registration
Embedded and edge	NVIDIA Jetson AGX Xavier, JetPack, Intel Movidius Myriad VPU, Intel Core i7 industrial edge (VTC7252-7C4IP), embedded Linux, TensorRT, OpenVINO, ONNX, latency and throughput profiling, Intel VTune
Software engineering	Linux, Git, branching, code review, CMake, Boost, Docker, containerised deployment, CI/CD, gtest, pytest, unittest, integration testing, multiprocessing, REST API design, modular design, debugging
Reliability	Structured logging, health checks, watchdog services, observability, production monitoring, drift tracking, failure-mode analysis, technical documentation, Apache Kafka, PostgreSQL
Computer vision	Object detection, object tracking, optical flow, segmentation, PyTorch, OpenCV, NumPy, YOLO, ByteTrack, NVIDIA DeepStream, GStreamer, FFmpeg
Professional	Agile and Scrum, Jira, mentoring, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage – Technical Assessor, Computer Vision and Software Engineering 05/2026 – PRESENT · REMOTE
Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

Taco Bell – Service Champion / Team Trainer 09/2024 – PRESENT · BRISTOL, ENGLAND
Progressed to Team Trainer in a customer-facing environment, onboarding new staff and handling escalations.